SPORTS & ARTHROSCOPY · NEET-SS / FET / INI-SS

Rotator cuff

1. Anatomy

Muscle	Nerve (root)	Insertion	Action	Test
Supraspinatus	**Suprascapular (C5-6)**	Greater tuberosity (superior facet)	**Abduction initiation**, depresses/compresses the head	**Jobe (empty can) test**, drop arm test
Infraspinatus	**Suprascapular (C5-6)**	Greater tuberosity (middle facet)	**External rotation (with arm at side)**	**External rotation lag sign**, resisted ER
Teres minor	**Axillary (C5-6)**	Greater tuberosity (inferior facet)	**External rotation in abduction**	**Hornblower's sign** (ER in 90 degrees abduction), **drop sign**
Subscapularis	**Upper and lower subscapular (C5-6)**	**Lesser tuberosity**	**Internal rotation**, anterior stability	**Lift-off (Gerber)**, **belly-press (Napoleon)**, **bear-hug** (most sensitive for upper subscapularis), **internal rotation lag sign**

- **Rotator cable** (crescent-shaped thickening perpendicular to the cuff fibres, from the biceps to the infraspinatus insertion): **"suspension bridge"** load transfer: **crescent tears** (within the cable) can remain functional.

- **Footprint:** supraspinatus about **12-25 mm** medial-lateral; **"critical zone" (Codman)**: **hypovascular area about 1 cm proximal to the supraspinatus insertion** (controversial).
- **Force couples: coronal plane: deltoid vs inferior cuff (infraspinatus, teres minor, subscapularis)**; **transverse plane: subscapularis vs infraspinatus/teres minor**. Tears that **disrupt a force couple** (e.g. **posterosuperior tears extending into the infraspinatus and teres minor**) cause **pseudoparalysis**.
- **Long head of biceps (LHB):** from the **supraglenoid tubercle and superior labrum**; runs in the **bicipital groove** stabilised by the **biceps pulley (SGHL + CHL)** and **transverse humeral ligament**; **subscapularis tears are associated with medial LHB subluxation**.

2. Pathogenesis

- **Intrinsic (degenerative): age** (prevalence of asymptomatic full-thickness tears rises with age: **about 25% over 60 and about 50% over 80**), **hypovascularity**, **smoking**, **diabetes**, **hypercholesterolaemia**, genetic.
- **Extrinsic: subacromial impingement (Neer)**: **acromial morphology (Bigliani: type I flat, II curved (commonest), III hooked: associated with tears)**, **critical shoulder angle (CSA over 35 degrees: cuff tears; under 30 degrees: OA)**, **os acromiale**, **acromial spurs**; **internal impingement** (**throwers: posterosuperior glenoid contacts the undersurface of the posterior supraspinatus/anterior infraspinatus in abduction-external rotation**: **articular-sided partial tears + posterosuperior labral lesions**, associated with **GIRD**).
- **Traumatic tears** (younger patients, falls, dislocations in over 40s).

3. Classification

- **Partial-thickness tears: Ellman**: **A (articular), B (bursal), C (interstitial)**; **grade 1 under 3 mm, grade 2 3-6 mm, grade 3 over 6 mm (over 50% of tendon thickness, about 12 mm)**. **PASTA lesion** = **partial articular supraspinatus tendon avulsion** (commonest partial pattern).
- **Full-thickness tear size (DeOrio and Cofield): small under 1 cm, medium 1-3 cm, large 3-5 cm, massive over 5 cm**. **Gerber definition of massive tear: complete tear of two or more tendons**.

- **Tear shape (Burkhart/Ellman): crescent** (mobile medial-lateral: direct repair to bone), **U-shaped** (**margin convergence (side-to-side) first**, then repair to bone), **L-shaped**, **massive contracted immobile**.
- **Patte classification of retraction: stage 1** close to the insertion, **stage 2** at the humeral head, **stage 3** at the glenoid.
- **Goutallier classification of fatty infiltration (CT, adapted to MRI by Fuchs): 0 normal; 1 some fatty streaks; 2 more muscle than fat; 3 equal fat and muscle; 4 more fat than muscle**. **Grade 3-4 (over 50% fat): poor outcome, high re-tear, irreversible**.
- **Tangent sign** (supraspinatus does not cross a line drawn between the superior borders of the coracoid and the scapular spine on sagittal oblique MRI): **atrophy**.
- **Hamada classification** (cuff tear arthropathy; see shoulder arthroplasty).
- **Subscapularis tears (Lafosse):** I partial upper third; II complete upper third; III complete upper two-thirds; IV complete with head centred; V complete with eccentric head (subluxation).

4. Clinical evaluation

- **Pain** (night pain, lateral arm/deltoid region), **weakness**, **painful arc (60-120 degrees)**.
- **Impingement tests: Neer sign** (passive forward flexion with internal rotation) and **Neer test** (pain relieved by subacromial lidocaine injection), **Hawkins-Kennedy test** (forward flexion 90 degrees + internal rotation).
- **Strength tests:** see table; **external rotation lag sign** (infraspinatus), **hornblower (teres minor)**, **drop arm (Codman)**.
- **Pseudoparalysis**: **active forward elevation under 90 degrees with full passive range** (massive tears).
- **Imaging:**
  - **Radiographs: acromiohumeral interval (AHI under 7 mm suggests a massive cuff tear; under 5 mm = irreparable)**, **greater tuberosity sclerosis/cysts**, **acromial morphology (outlet view)**, **critical shoulder angle**, cuff tear arthropathy changes.
  - **Ultrasound** (dynamic, cheap; operator-dependent; similar accuracy to MRI for full-thickness tears).

- **MRI** (tear size, retraction, **atrophy/fatty infiltration**, LHB, labrum); **MR arthrography** (articular-sided partial tears).

5. Management

Subacromial pain (impingement) without full-thickness tear

- **Physiotherapy (scapular and cuff strengthening), NSAIDs, subacromial steroid injection.**
- **Subacromial decompression (acromioplasty): no better than placebo/arthroscopy only (CSAW, FIMPACT trials)**: limited role.

Partial-thickness tears

- **Non-operative first**; **under 50%**: **debridement ± acromioplasty**; **over 50% thickness (Ellman 3)** or bursal-sided over 25-30%: **repair** (**transtendon repair for PASTA** or **completion and repair**).

Full-thickness tears

- **Acute traumatic tears in active patients: early repair (within about 3-6 weeks)** (before retraction and fatty infiltration).
- **Chronic degenerative tears: trial of physiotherapy** (successful in **about 75%** in some studies), **repair if persistent symptoms**, especially **younger, active patients**, with **reparable tears (good muscle quality: Goutallier under 3)**.
- **Natural history of untreated full-thickness tears: about 50% enlarge** (over years; **symptomatic tears, larger tears and those involving the anterior cable are more likely to progress**).

Repair techniques

- **Arthroscopic** (standard) vs **mini-open** vs **open** (similar outcomes).
- **Single-row** vs **double-row** (**double-row: better footprint restoration, higher biomechanical strength, lower re-tear rates for larger tears**, but similar clinical outcomes overall) vs **transosseous-equivalent (suture-bridge)** (**most widely used for medium-large tears**).
- **Margin convergence** for U-shaped tears (reduces strain on the repair).
- **Biceps tenotomy or tenodesis** if the LHB is diseased/unstable (**tenotomy: Popeye deformity, cramping**; **tenodesis: younger/cosmetic concern**).

- **Rehabilitation: sling for about 4-6 weeks** (passive ROM early or delayed: **similar outcomes; early motion may increase re-tear in large tears**), active at about 6 weeks, strengthening at about **12 weeks**; tendon-bone healing about **12-16 weeks**.

Outcomes and re-tears

- **Re-tear rates: about 10-20% for small/medium tears, up to 40-90% for massive tears** (MRI); **re-tears often still have good pain relief**, though **healed repairs have better strength**.
- **Risk factors for re-tear: age over 65**, **large tear size**, **retraction**, **fatty infiltration (Goutallier 2+)**, **smoking**, **diabetes**, osteoporosis, **delayed repair**, poor tendon quality, **workers' compensation**.
- **Re-tear pattern: Cho type 1 (at the footprint: commonest with single-row)** and **type 2 (at the musculotendinous junction, medial to the medial row: double-row)**.

6. Massive irreparable cuff tears

- **Options (age, pseudoparalysis, arthritis, subscapularis and teres minor status):**
  - **Non-operative**: deltoid rehabilitation (anterior deltoid programme).
  - **Arthroscopic debridement ± biceps tenotomy** (pain relief in low-demand elderly; **preserve the coracoacromial arch**: **avoid acromioplasty/CA ligament release** to prevent **anterosuperior escape**).
  - **Partial repair** (restore force couples: repair infraspinatus/subscapularis).
  - **Superior capsular reconstruction (SCR)** (**Mihata**: **fascia lata autograft (thick, 6-8 mm)** or **dermal allograft** from the superior glenoid to the greater tuberosity: restores **superior stability**; **young patients without arthritis, intact/repairable subscapularis**).
  - **Subacromial balloon spacer** (InSpace: **no benefit over debridement in START:REACTS trial**).
  - **Tendon transfers:**
  - **Posterosuperior tears (loss of ER): latissimus dorsi transfer (Gerber)** (**contraindicated with subscapularis deficiency**), **lower trapezius transfer (Elhassan; with Achilles allograft)** (more in-phase with ER).

- **Anterosuperior/subscapularis tears: pectoralis major transfer** (**sternal head passed under the clavicular head (behind the conjoint tendon: more anatomic vector)**), **latissimus dorsi anterior transfer**.
- **Reverse total shoulder arthroplasty** (**elderly, pseudoparalysis, cuff tear arthropathy**).
- **Interposition grafts/patch augmentation**.

7. Related conditions

- **Calcific tendinitis: supraspinatus (commonest)**, **women aged 30-50**, **hydroxyapatite (calcium hydroxyapatite) deposition**; **not degenerative** (cell-mediated). **Uhthoff stages: precalcific (fibrocartilaginous metaplasia)**, **calcific (formative phase, resting phase, resorptive phase: most painful, acute; deposit fluffy/ill-defined)**, **postcalcific (reconstitution)**. **Gärtner classification** (type I well-defined dense; II mixed; III cloudy translucent: resorptive). **Treatment: NSAIDs, physio**, **ultrasound-guided barbotage (lavage)** (best evidence), **extracorporeal shock wave therapy (ESWT)**, **arthroscopic excision**.
- **Frozen shoulder (adhesive capsulitis): diabetes (strongest association; about 10-20% of diabetics; worse prognosis)**, thyroid disease, Dupuytren, Parkinson, post-surgery, **women aged 40-60**. **Loss of active and passive ER (with arm at side) is the hallmark**; normal radiographs. **Phases: freezing (painful), frozen (stiff), thawing** (**about 1-3 years**). **Pathology: fibroblast/myofibroblast proliferation** (**Dupuytren-like**, type III collagen) in the **rotator interval and CHL**. **Treatment: intra-articular steroid injection (best early evidence)**, physiotherapy, **hydrodilatation**, **manipulation under anaesthesia (MUA)**, **arthroscopic capsular release** (**UK FROST trial**: all three (early structured physio + steroid, MUA, ACR) improved similarly; ACR more costly/risky).
- **SLAP lesions (superior labrum anterior to posterior) (Snyder): I** degenerative fraying (debride); **II superior labrum and biceps anchor detached** (**commonest**: repair in young overhead athletes; **biceps tenodesis in over 35-40 years**); **III bucket-handle tear of the labrum, biceps intact** (excise); **IV** bucket-handle extending into the biceps (repair/tenodesis depending on biceps involvement). **Tests: O'Brien (active compression)**, **Speed's**, **dynamic labral shear (O'Driscoll)**. **Spinoglenoid**

paralabral cyst (suprascapular nerve compression: infraspinatus weakness).

- **Subscapularis tears and LHB instability**: **medial subluxation of the LHB** suggests a subscapularis tear.
- **Os acromiale: failure of fusion of acromial ossification centres (meso-acromion commonest)**; **fuse by about 25 years**; can cause impingement; **ORIF with bone graft** if symptomatic.

> Exam pearl: **Supraspinatus/infraspinatus = suprascapular nerve; teres minor = axillary; subscapularis = upper/lower subscapular. Belly-press/bear-hug: subscapularis; hornblower: teres minor; ER lag: infraspinatus.**

> Exam pearl: **Goutallier 3-4 (fat equal/greater than muscle) = irreversible, high re-tear. AHI under 7 mm = massive tear. CSA over 35 degrees = cuff tear. Acromioplasty no better than placebo (CSAW). Re-tear risk: age, size, retraction, fatty infiltration, smoking.**

> Exam pearl: **Massive irreparable tears: SCR (young, no OA), LD or lower trapezius transfer for posterosuperior; pectoralis major transfer for subscapularis; RTSA for elderly pseudoparalysis/CTA. Preserve the CA arch (anterosuperior escape). Frozen shoulder: diabetes; loss of passive ER; FROST trial. SLAP II commonest (repair young; tenodesis older).**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.